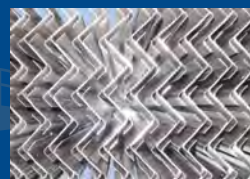
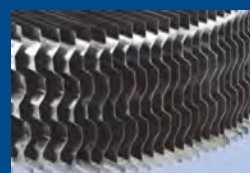




SEPARATION TECHNOLOGY



INNOVATIVE SOLUTIONS **RELIABLE PERFORMANCE**

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Your Partner in Separation Technology

GTI Solutions is a process equipment and technology company committed to providing innovative mass transfer & separation equipment solutions to the process industries.

Our expertise in process design, engineering, and manufacturing allow us to deliver the most cost effective design for all your mass transfer and separation applications.

Our mass transfer and separation equipment are tailor-made to operate in a wide range of applications and can be fabricated in a variety of corrosion resistant materials such as stainless steel and other high alloys metals plus all types of plastics, including engineered plastics.

The application expertise of our engineers, our product design experience and engineering knowledge coupled with extensive range of conventional and propriety products allow us to design the most optimum solutions for any of your gas/liquid and/or liquid/liquid separation issues with two or three phase separators.

With a complete portfolio of mass transfer and separation products, GTI Solutions offer a SINGLE WINDOW for all your process engineering needs. Our portfolio of products and services includes:

- Conventional trays
- High performance trays
- Random packing
- Structured packing
- Grid packing
- Column internals
- Mist eliminators (mesh & vanes)
- Liquid-liquid coalescers
- Reactor internals
- Inlet defoaming cyclone bundles
- Gas outlet cyclone tube array
- Divided wall column technology
- Process validation through simulation
- Unique retrofit technology
- CFD modeling
- Engineering services
- Feasibility and complex revamp studies
- Expedited replacement services
- Troubleshooting services & onsite services
- Technical training
- Site supervision services

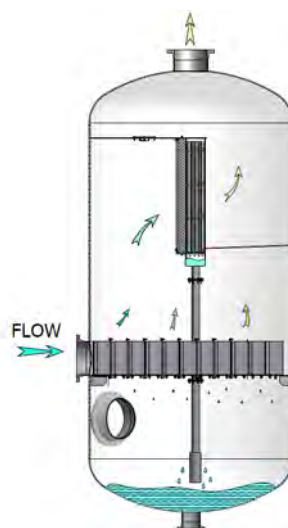
Entrained liquid, when not properly removed from gas stream, can lead to various operating issue including but not limited to contamination of products, corrosion of downstream equipment, high solvent losses, catalyst poisoning and/or equipment damage. GTI Solutions Mist Eliminators in such situations become an efficient solution by removing entrained liquid, thereby ensuring trouble free operation

Mist Eliminators come typically either as knitted wire mesh pad type or mesh vane assemblies. The type of mist eliminator to be employed is basically a function of:

- Liquid droplet size being encountered
- Removal efficiency required
- Capacity requirement
- Nature of fluid being handled such as fouling tendency and possibility of entrained solids
- Operating pressure
- Or client given separator limitations

GTI Solutions also provides G-MAXSEPT™ Mist Eliminators that uses two stage mist eliminations solutions to maximizes both capacity and separation efficiency.

Liquid-liquid coalescers are used to separate two immiscible liquid phases where gravity separation is not sufficient to reach a desired separation efficiency and/or results in large vessel size. In this scenario, GTI Solutions employs knitted mesh packs or plate packs and proprietary cylindrical coalescers as coalescing media. With a broad selection of products, unmatched application expertise and excellent process engineering knowledge, allow GTI Solutions to provide the most optimum solution even for the most challenging emulsified separations, for example hazy jet fuel, caustic entrainment of gasoline, or LPG or oil contamination in the sour water stripper feed.



Mesh Pad Mist Eliminator Technology

GTI Solutions mesh pads, G-M™ Wire-Mesh mist eliminators, are knitted mesh pad that are manufactured from woven wires that is crimped and formed into a flat pad and fitted into the tower or separator. The crimped mesh layers are sandwiched between top and bottom grids made from metal bars and/or rods.

G-M Wire-Mesh mist eliminators function primarily on inertia impaction and are efficient in removing droplets 1 μ or larger, depending upon process conditions and applications. Layers of crimped mesh provide large surface area in a small volume ensuring good liquid removal without restricting gas flow. Mist eliminators coalesce small liquid droplets into larger ones to allow easy drainage by gravity force. G-M mist eliminators are typically installed horizontally in vessel with gas flowing upward. However, for certain special application, including compact separation vessels, G-M mist eliminators can also be installed vertically for horizontal gas flow.

G-M mist eliminators can be made from metal and non-metal wires such as synthetic fibers or plastics, depending on applications or client requirements. For certain applications that require very fine separation, co-knit mesh pad can be applied to maximize de-entrainment capability. Here metal + polymer or glass fiber wires are co-knitted together to maximize separation efficiency.

SELECTION OF G-M MIST ELIMINATORS

Selection of mist eliminator is a function of:

- Gas & liquid flow rates and viscosity
- Expected liquid entrainment
- Location where the mist eliminator needs to be installed in the vessel
- Allowable pressure drop
- Level of efficiency to be reached
- Operating pressure of the application

GTI Solutions provides a wide range of mist eliminators, including engineered mist eliminators, to meet every process and application needs. Some of the common mist eliminator are listed in Table-1 below.

Table 1. Wire Mesh Mist Eliminator Styles

Style	Density, kg/m ³	Free Volume, %	Surface Area, m ² /m ³	Characteristics
G-M 139	80	99.0	158	Medium Efficiency
G-M 139H	85	99.2	155	Medium Capacity
G-M 163	120	98.5	210	Ultra High Efficiency
G-M 491	126	98.6	455	High Efficiency / High Capacity
G-M 623	128	98.4	460	High Efficiency
G-M 134	144	98.2	280	General Purpose
G-M 197	146	98.4	278	High Efficiency / High Capacity
G-M 173	155	94.0	520	Ultra High Efficiency
G-M 512	160	94.0	535	Ultra High Efficiency / High Capacity
G-M 1117	170	98.0	355	High Efficiency / High Capacity
G-M 124	173	97.7	360	Heavy Duty / High Efficiency
G-M 214	193	97.6	375	High Efficiency
G-M 512c	Ultra-Efficiency and proprietary design (please contact GTI Solutions for more details)			
G-M 182c				

* Additional styles are available upon request

Mist Eliminator Sizing

For mist eliminator to work effectively the key variable is the gas velocity, whereas a given application there is a defined operating range. To determine the required mist eliminator diameter and therefore size of the vessel, maximum acceptable superficial velocity should be established using the Souders-Brown equation.

$$Vd = k \sqrt{\frac{\rho L - \rho V}{\rho V}}$$

Where:

- Vd : Maximum design velocity
- ρV : Gas density
- ρL : Liquid density
- k : Capacity-factor

In general, capacity factor (k) is a function of type & style of separation equipment used.

Table-2 provides typical K factor that are applied for different type of Mist Eliminator Design using general design guidelines. Care should be taken to derate and or adjust K factor for different operating including but not limited to,

- Operating pressure
- Gas & liquid viscosity
- Liquid loading and
- Surface tension

Table 2. Typical allowable k-factors and operating range for different type of mist eliminators

Type	Allowable k-factor	
	(m/s)	(ft/s)
No Mist Eliminator (gravity separation)	0.07	0.23
Standard Mesh Pad	0.11	0.35
High Capacity Mesh Pad	0.12	0.40
Standard Vane type mist eliminator	0.15	0.50
High Capacity Vane type mist eliminator	0.33	1.10
G-MAXSEPT™ Separation Systems	0.35	1.15



Note: No matter which separation internals are used, the maximum operating range is 30% ~ 110%.

Use of correct k-factor for sizing mist eliminator is critical for its performance and it is recommended that you contact a GTIS engineer before finalizing the design.

The ultimate selection of type of mist eliminator is not only based on capacity requirement but also a function of separation efficiency, fouling tendency of the fluid being handled and allowable pressure drops across the equipment.

Advantage of G-M Mist Eliminators

- Ease of installation
- High separation efficiency with low pressure drop
- Most economical solutions for a given performance
- Available in wide varieties of material and densities to meet specific process requirements

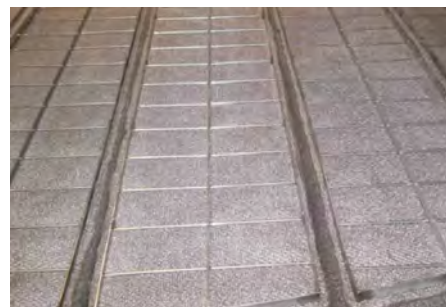
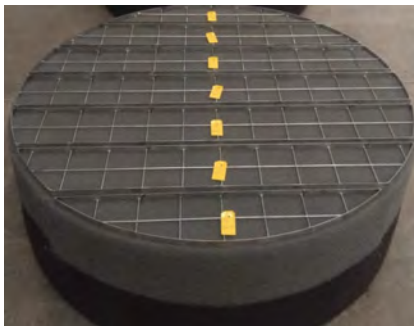
Engineered Mist Eliminator, Style G-M 512c and G-M 182c

Co-Knit mesh pads are high efficiency mist eliminators especially designed to enhance the separation performance. They consist of a mixture of metal and non-metallic wires such as glass, teflon or plastic fibres. The internal geometry is tailor-designed to promote droplet size growth making it easier for separation while ensuring maximum liquid throughput velocities for minimum vessel sizes.

G-M 512c is made by combining metal wire with glass fibre while G-M 182c is a co-knit of metal plus PTFE. Both mesh pads provide much higher efficiency when compared to standard single (metal) wire mesh pad. They are also preferred coalescing medias enhancing separation of hydrocarbons from solvents and/or water.

Some typical applications of G-M engineered mist eliminator include TEG Contactors and amine sweetening units. Use of G-M engineered mist eliminators at the top of these contactors can allow operators to reduce solvent losses to as low as 0.1 gpm/MMSCFD, leading to significant saving in operating cost due to lower solvent losses.

G-M engineered mist eliminators are also installed upstream of these Contactors to minimize solid particles carryovers into the Contactor that could otherwise induce foaming while limiting its capacity and efficiency.



Vane Type Mist Eliminator Technology

G-C Chevron mist eliminators are high capacity separators manufactured from a set of parallel vane plates arranged in a pre-determined geometry as a function of capacity and required performance.

G-C Chevron mist eliminators function primarily on inertial impaction and are efficient in removing particles 8μ or larger, depending upon process conditions and applications. The geometry of the vanes causes gas to change directions multiple times as it passes through the mist eliminator. Entrained liquid droplets cannot follow the gas streamlines and impinge upon the walls of the vanes, coalesce to a larger droplet size where they rain down due to gravity and are removed.

G-C Chevron mist eliminators are preferred over mesh type mist eliminators when there is a concern of fouling, including solid carryover or when the entrained liquid is highly viscous. G-C Chevron mist eliminators can be designed for either vertical or horizontal flow based on vessel orientation or application requirements. Vanes come in 2, 4 and 6 passes based on performance required from the mist eliminator or fouling potentials of the application.

G-C Chevron mist eliminators can be made from metal and non-metal (plastics or FRP), depending on applications or client requirements. For certain applications that require greater efficiency or higher capacity, a more complex vane design can be employed using separate channels for liquid disengagement and drainage from the vane surface.

SELECTION OF G-C Chevron MIST ELIMINATORS

The type of G-C Chevron mist eliminator to be selected is a function of:

- Gas and liquid flow rates
- Required performance
- Location in the vessel where mist eliminator is installed
- Applications
- Physical and chemical properties of fluids being handled
- Material of construction



Typical applications of G-C Chevron mist eliminators are:

- Viscous or sticky liquid that can plug knitted mesh pad mist eliminators
- Systems prone to fouling or gas stream carrying solid particles along with entrained liquid
- Application requiring wide operating range or low pressure drops

HIGH CAPACITY G-C CHEVRON MIST ELIMINATOR

Conventional chevron mist eliminator uses vane surface for impingement, coalescence and drainage of liquid with no shielding of captured liquid from the up flowing vapor. At higher gas flow rates, this can lead to re-entrainment and compromising separation efficiency.

High capacity G-C Chevron Mist Eliminator uses vanes that are equipped with liquid capture pockets on either side of the wall, which act as drainage channels for the coalesced / capture liquid, shielding it from the vapor flow and thereby avoiding re-entrainment. This makes them an ideal choice to debottleneck existing equipment or reduce vessel size for new applications.



G-C Chevron Mist Eliminator (Double Pocket Vane)

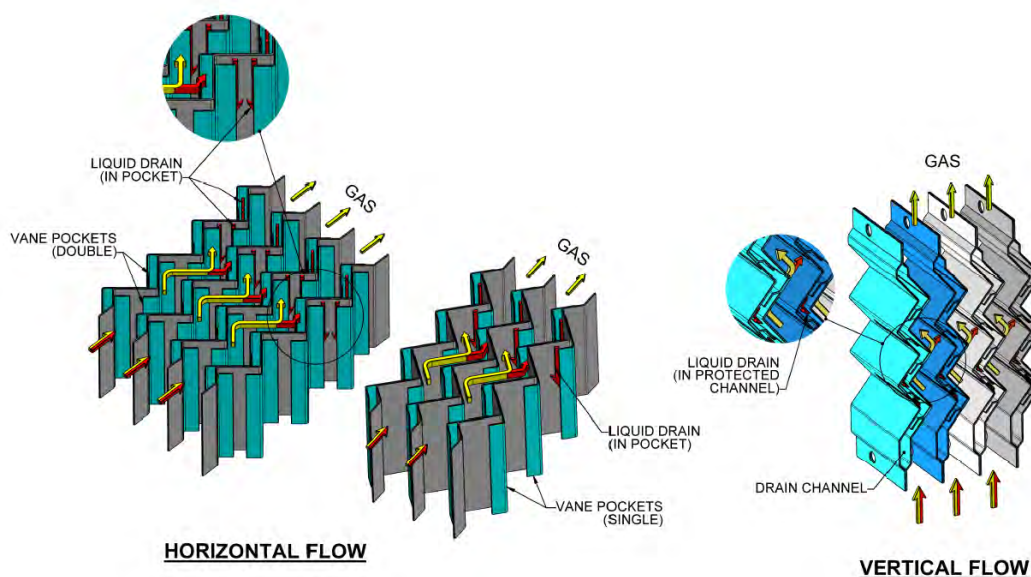
GTI Solutions provides a wide range of chevron mist eliminator solutions, including engineered mist eliminators, to meet every process and application needs. Table 3 provides a list of widely used G-C Chevron mist eliminators.

Table 3. G-C Chevron Mist Eliminators Styles

Style	Description
G-C 201	2-pass Chevron Vane Mist Eliminator
G-C 301	3-pass Chevron Vane Mist Eliminator
G-C 401	4-pass Chevron Vane Mist Eliminator
G-C 601	6-pass Chevron Vane Mist Eliminator
G-C 151SP	High Capacity Single Pocket Vane Mist Eliminator
G-C 251DP	Horizontal Flow High Capacity Double Pocket Vane Mist Eliminator
G-C 351DP	Vertical Flow High Capacity Double Pocket Vane Mist Eliminator

* Additional styles are available upon request

G-C-Chevron Mist Eliminator's Design & Function



Liquid-Liquid Coalescer

GTI Solutions utilizes latest technology to offer clients the most effective liquid-liquid coalescer that separate a wide range of two immiscible liquids. Products that are emulsified and hazy are very difficult to separate, hence they require a special coalescer design.

The environmental policies have become increasingly stringent in response to environmental challenges. Policies imposed by governments around the world have implemented regulation that forces the industry to seek assistance of liquid- liquid coalescing technology.

When retrofitting existing liquid-liquid separators for up to 100 % more capacity and/or improved efficiency, GTI Solutions can design and manufacture coalescer internals, including vessel.

Radial flow coalescers are the latest technology available, which allow to design high capacity, and/or improved efficiency applications while the vessel size remains small. Radial flow coalescers are also the preferred choice with the complete skid mounted 2 or 3 phase coalescer systems.

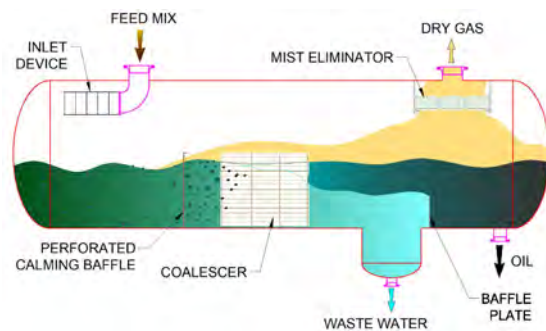
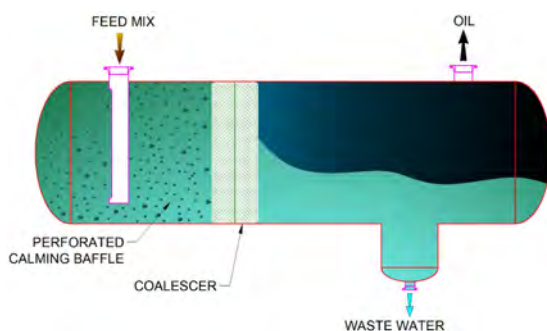
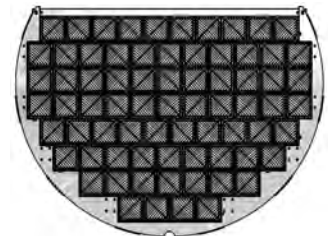
GTI Solutions can provide efficiencies of <15 ppm of free water in oil or free oil in water. If solids are present, the internals can be plugged. Traditionally, a dual prefilter skid is used to remove any solids before entering the horizontal coalescer separator. To optimize the overall design, GTIS includes a high efficiency plate pack as the 1st stage to remove solids greater the 10 micron as well as larger liquid droplets.

Now the coalescer pack becomes the 2nd stage in the same horizontal separator, and will remove free water in oil or free oil in water down to < 15 PPM. GTIS solutions is a more cost be effective design.

The coalescer will remove solids greater than 10 microns and is installed in the same coalescer vessel as a 1st stage which is a more cost effective solution.

Advantages of Liquid-Liquid Coalescer

- Reduces cost due to smaller vessel design
- Capacity increase for the existing vessels
- Higher recovery of valuable products
- Removal of haziness in the product
- Technology to remove the most difficult emulsions
- Reduces waste water production from tankage, separator and desalter

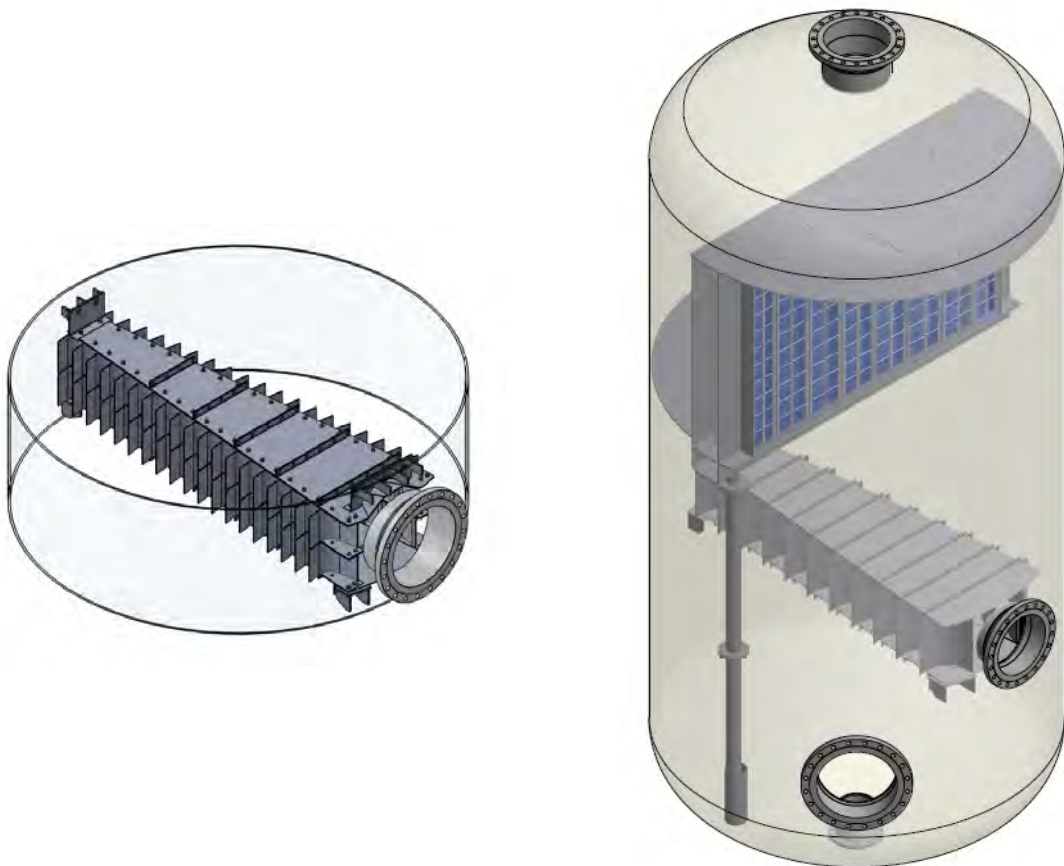


Inlet Diffuser : G-Vane™ 600 Multi-Vane Distributor

Utilization of Inlet diffusers are often overlooked but yet an important device for designing efficient separators for the industry. G-VANE 600 Multi-Vane Distributor is designed to enhance the capacity and efficiency of separators 2-3 phases and for horizontal and vertical vessels. G-Vane can be manufactured with any types of metal.

Advantages of G-Vane 600 Multi-Vane Distributor

- Reduces high inlet velocity from the incoming gas thus improves the downstream separation equipment efficiency
- Dissipates high inlet gas velocities efficiently, avoiding localized high velocities
- Evenly distributes the gas flow throughout the separator with minimal pressure drops
- Prevents inlet gas to jet into the bottom liquid level and re-entrain additional liquid droplets that may potentially flood the downstream mist eliminator
- Reduces vessel height or length
- Allows higher gas flow in a smaller separator without creating localized channeling intern
- Removes mild foaming, if present at the inlet gas stream



Filter Separator

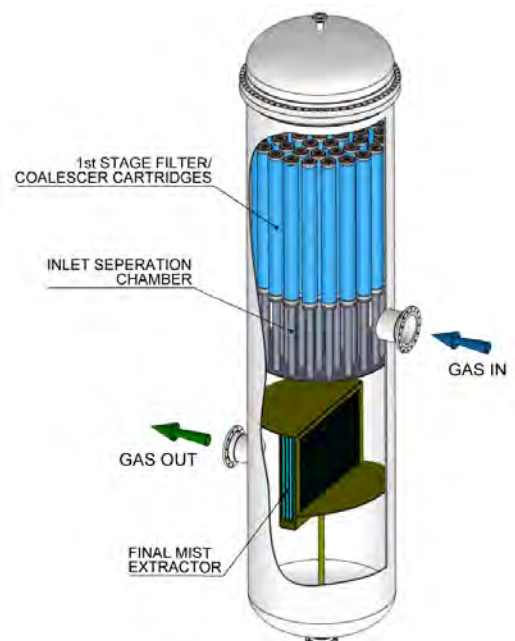
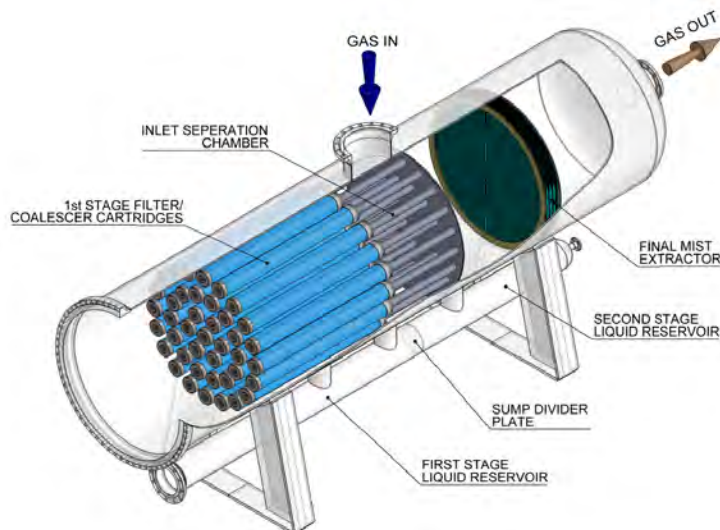
GTI Solutions' two stage filter separator technology will achieve the smallest possible vessel diameter, thus resulting in the lowest capital investments while removing aerosol liquids and solid particles efficiently.

The first stage contains replaceable coalescing filters mounted on a tube sheet. Here the bulk liquids and solids are removed while coalescing the aerosol liquid droplets into larger ones (greater than 10 microns).

Some of these coalesced droplets may fall off the filter but the majority is pushed to the second stage due to the high surface velocity. In the second stage, these large droplets are removed by a high capacity and high efficiency Double Pocket Vane, such as G-C-251_{DP} & G-C-351_{DP}. Filter separators can be designed both vertically and horizontally.

Advantages of Filter Separator

- Removes aerosol mist to protect downstream equipment from erosion, fouling, pitting corrosion or deactivating of catalyst or molecular sieves
- Reduces amine or glycol foaming problems by removing solids and free liquid particles from the inlet gas stream
- Allows low cost vessel designs
- Debottleneck existing equipment by adding/retrofitting high capacity double pocket vane as a 2nd stage we allow for more capacity without requiring a new separator
- Protects ultra-low NOx burners from plugging, therefore preventing furnace heating imbalance



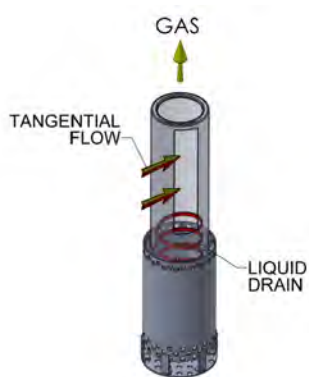
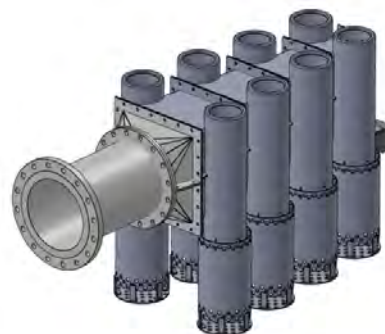
Inlet Cyclone Separator Cluster

The cyclonic inlet device is configured for tangential entry of the fluid into the vessel to create centrifugal acceleration for improved separation of gas and liquid. The cyclone in form of a cluster made of multiple cyclones, can be used in vertical or horizontal vessels intended for gas or liquid dominant flow. There are several benefits of incorporating cyclone cluster

The cyclone separator consists of multiple vertical cylindrical tubes internally mounted and either welded or flanged to the inlet nozzle of the separator. When specified, the cluster can be supplied as unassembled bolted tubes so that the tubes can go through the manway of the vessel and the cluster can be assembled inside thereafter.

Advantages of Cyclone Separator Cluster

- Eliminates Foam
- Streamline process fluid entry
- Accelerate separation process
- Improves separation efficiency
- Reduces retention time and size of the vessel



The process fluid enters the cyclonic device through a common header specifically engineered to uniformly distribute the flow in each cyclone tube.

The flow is then directed into the vertical tubes tangentially which causes the process fluid to undergo centrifugal acceleration. The force caused by the vortex movement enhances the separation of heavier liquid and/or solid particles from inlet stream.

The cyclonic separation occurs at much faster rate than it would in a convention separator, hence reducing retention time and size of the vessel.

G-MAXSEP™ High Performance Mist Eliminators Technology

Vertical knock-out drums such as those in ethylene plant compressor trains and/or LNG plants, the separator size is established by capacity of mist elimination devices. Mesh pads can provide good separation efficiency, but depending of the operating conditions, it might require a much larger vertical KO drum diameter and therefore increasing the project cost. GTIS offers a unique design called G-MAXSEP which consists of 2 stage mist eliminator system. It provides the extra capacity and increased efficiency in compactor vertical KO drum design.

G-MAXSEP high performance mist eliminators technology overcome this limitation by using two-stage mist eliminations. Here we employ a combination of mesh pad and double-pocket vane mist eliminator, benefiting from high efficiency of mesh pad and high capacity of double-pocket vane mist eliminator.

The mesh pad is placed upstream of the double-pocket vane and act as an agglomerator, coalescing tiny entrained liquid particles into large particles that are eventually removed by downstream double-pocket vane mist eliminator.

Table 4. Performance of G-MAXSEP High Performance Mist Eliminator

Style	Performance
G-C 201	~99% Removal of droplets > 40 μ
G-C 401	~99% Removal of droplets > 25 μ
G-C 601	~99% Removal of droplets > 15~20 μ
G-C 151SP	~99.5% Removal of droplets > 10 μ
G-C 351DP / 251DP	~99.9% Removal for ~8 μ liquid droplet size
G-MAXSEP	~99.9% Removal for ~1 μ liquid droplet size

The use of two stage mist elimination allows G-MAXSEP to be excellent revamp tool, boosting capacity increase, up to 3 times, when compared to conventional mist eliminators. Existing vessels can be revamped with G-MAXSEP internals without any hot-work (welding) within existing vessel, allowing operators to achieve revamp objectives with minimum cost and or shutdown period.

For new vessel use of G-MAXSEP, can allow the vessel size to be reduced by 25%-40%, providing significant reduction in capital cost especially for LNG plants that use thicker wall vessels due to high operating pressure.

The G-MAXSEP solution also provide higher operating range when compared to using just a single mist eliminator (typically 10~110% range). At low gas flow, standalone mist eliminator have poor separation performance. In case of G-MAXSEP internals, the 1st stage will act primarily as a mist eliminator at low flow rate, providing required separation performance.

Alternative design can be made with the mesh pad located downstream of the chevron mist eliminator when high liquid handling carrying capacity is required. In this design, the chevron mist eliminator does the coarse separation, removing larger entrained particles, while the mesh pad does the fine separation de-entraining fine or smaller particles thereby maximizing the efficiency of entire assembly.

G-MAXSEP separator internals in conjunction with G-VANE 600 Multi Vane Distributor will reduce the high momentum of incoming feed in a controlled manner with minimum pressure drop. This minimizes liquid droplet atomization, thereby avoiding additional entrainment.

In addition, the guide vanes uniformly distribute the feed in multiple horizontal streams while reducing the velocity, and improving vapor distribution.

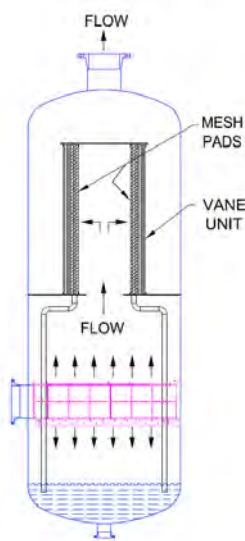
Benefits of G-MAXSEP

- Increase capacity up to 300%
- Reduce new vessel size and weight
- Increase separation efficiency
- Increased operating range, typically 10~100%
- Debottleneck existing equipment
- Avoids direct welding to vessel walls if using Quick-Fit solution in retrofit application

Separation Technology To Retrofit Traditional K.O. Drums

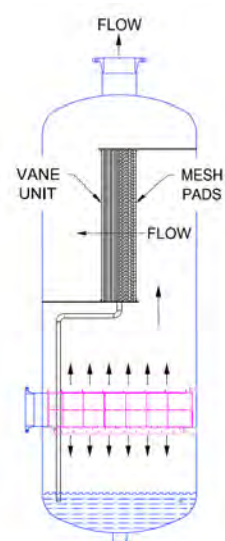
Retrofit with high efficiency / capacity 2 stage mist eliminators providing a one or two bank arrangement, based on the gas flow rate and the available spaces in the K. O. Drum. In most cases, an inlet diffuser is also required.

This solution also provide high operating range 10~100% , when compared to using just a single mist eliminator. At low gas flow standalone mist eliminator have poor separation performance. In case of G-MAXSEP systems, the upstream unit will act as primary mist eliminator at the low flow rate condition, providing the required separation performance.



Generally, the optimum mist eliminator performance in terms of both efficiency and capacity, is achieved by providing a special double pocket vane downstream of a mesh agglomerator using a horizontal flow.

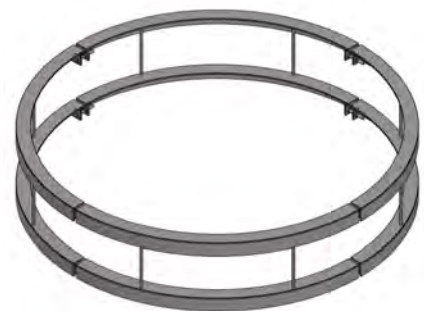
The mesh agglomerates fine mist droplets into larger ones that are subsequently captured by the double pocket vane(s). The special double pocket vanes shield the captured liquids from being re-entrained.



Retrofit without recertifying for ASME code using QUICK-FIT™

Retrofitting an existing vessel often has one drawback: it often requires welding new support rings, beams, clips or other structures to the vessel wall. Most vessels are ASME code certified, thus, after welding to the vessel wall, the welded area must be heat-treated, and the vessel must be recertified. It is generally desirable to avoid this cumbersome procedure.

The QUICK-FIT is a solution to avoid direct welding to vessel walls. Expansion rings are bolted together and wedged against the vessel wall without welding. The unique double expansion design ensures that the installed rings are a tight-fit and do not move during operation. Once the rings are installed, either in a vertical or horizontal vessel, beams or pedestal supports can be bolted or welded to the rings and complete housings can be built inside the vessel.



QUICK-FIT expansion rings for retrofitting knockout drum internals without welding

G-Cyclonic Mist Eliminator Technology

Both mesh pads and vane type mist eliminators work as gravity separators for gas-liquid separation. When dealing with high pressure (> 1000 psig) application such as separation drum in gas processing or LNG plants, the capacity of these devices is limited due to low liquid to gas density ratios. Use of G-cyclonic mist eliminators employs centrifugal force to achieve separation, thereby not limited by gravity separation.

The main elements of the axial flow cyclone are a series of cyclone tubes with swirl-body inside the tubes. The gas flows through these tubes and the blades of the swirl-body in the cyclone brings the gas flow into a swirling motion which causes the liquid droplets in the gas stream to migrate to the cyclone tube wall, due to the centrifugal forces generated by this swirling motion.

Eventually, the droplets will be caught by the tube wall and will form a liquid film that will spiral upwards on the inner tube wall. Slots are provided in the tube wall to transport the "separated-off" liquid to the liquid collection chamber of the multi-cyclone bundle and eventually via a liquid drain to the liquid compartment of the gas/liquid separation vessel.



This design arrangement ensures maximum de-entrainment of liquid particles exiting the cyclone assembly. The number of swirl elements can be optimized based on vessel diameter, required capacity & performance, ensuring lowest cost per unit rate of gas flow being handled.

As G-Cyclonic Mist eliminator are not limited by gravity separation, this technology is capable of processing extremely high gas flow capacity, up to 300%+ when compared to mesh pad and or chevron mist eliminators. This allow our engineers to significantly reduce vessel diameter and height. For high pressure separators such as those in FPSO, this leads to lowest vessel foot print, significantly reducing overall construction cost. For revamp G-Cyclonic can provide additional gas handling capacity that may not be available with other mist elimination technologies.

G-MAXSEP system can also be applied to G-Cyclonic device by combining G-VANE 600 inlet device and mesh pad agglomerator. The vane inlet device ensures entry of fluid inside the vessel controlled momentum which will impart uniform distribution across entire vessel cross section. The vapor, along with any entrain liquid passes through the mesh pad that's work as an agglomerator, coalescing small entrained liquid particles in to large particles. The mist laden vapor will then pass through the cyclonic separators that provide the required separation ensuring mist free gas leaving the separators. For applications that are prone to fouling, the mesh pad agglomerator can be replaced by G-C Chevron mist eliminator to ensure trouble free operation.

Benefits of Cyclone Mist Eliminators

- High gas handling capacity (vessel gas K value up to < 0.5 m/s)
- High liquid removal efficiency ($> 99\%$ droplets larger than 10μ)
- High turndown ratio (factor of 10)
- Ability to handle slugs
- Pressure drops varies from 0.3~0.5 PSI (18~34 mbar)
- Minimize separator size especially the OD
- Ideal for capacity retrofits in existing separators installed via the manway
- Removes solids

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